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THE UNITED REPUBLIC OF TANZANIA

PRESIDENTS' OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

SINGIDA REGION

FORM FOUR MOCK EXAMINATION

032/1 CHEMISTRY 1

Time 3:00 Hours 21st July, 2021 a.m

1. Instructions

2. This paper consists of section A B and C
3. Answer all questions in section A and B and one (1) question from section C
4. Where necessary the following constants may be use;

Atomic masses
H =1, O=16, N=14, S=32, C=12, Na=23, Pb=207, Cu=63.5, Zn=65,
Avogadro's S.T.P = 22.4dm^3
1 Faraday = 96,500coulomba
1 Litre = 1dm^3 = 1000cm^3

SECTION A (15 Marks)
Answer all questions in this section.

1. Choose the correct answer from among the given alternatives and write its letter beside the item number in the answer book provided.

(i)An industrial preparation of ethane from higher alkane is done by

A. Sublimation
B. Cracking
C. Distillation
D. Synthesis
E. Ethenelyzation

(ii) In ammonia gas, hydrogen and nitrogen atoms are held together by;

A. Covalent bond
B. Weak force of attraction
C. Ionic bond
D. Covalent
E. Ionization bond

(iii) The volume occupied by 8.4g of nitrogen gas at S.T.P is;

A. 22.4 dm3
B. 22400 dm3
C. 13.44 dm3
D. 6720 dm3
E. 67.20

(iv) When an atom loses an electron, it becomes;

A. Molecules
B. Isotopes
C. Cation
D, Proton
E. Anion

(v) A quantity of electricity is measured in;

A. Ampere
B. Watts
C. Coulombs
D. Seconds
E. Joules

(vi) Any substance produced and released into the environment that cause ugly to it is

A. Waste
B. Pollutants
C, Pollution
D. Polluter
E. Sewage

(vii)existence of two or more atoms with the same atomic number but different mass number is called;

A. Isotopy
B. Isotopes
C. Allotropes
D. Isobars
E. Allotropy

(viii) How can you obtain iron fillings from the mixture of solid iodine and iron fillings;

A. evaporation
B. fractional distillation
C. chromatography
D. decantation
E. sublimation

(ix) The primary macro elements needed by plants for proper growth and development are;

A. N, S and ca
B. P, N and C
C. N, P and K
E. N, Mg and Mn

(x) A piece of land has a soil with pH of 6.2, this indicates that the soil is;

A. High acidic
B. Strong basic
C. Both acidic and neutral
D. Slightly acidic
E. Slightly basic

2. Match the items in list A with the corresponding item in list B by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A LIST A
(i)Hydrogen sulphide .....
(ii) Allotrope of sulphur.....
(iii) Manufacturing of concentrated sulphuric acid .....
(iv) Used for making painter ink .....
(v) It decolorized acidified potassium permanganate .....

LIST A LIST B
A. Rhombic
B.sulphur dioxide
C.Sodium sulphate
D.Smell of rotten egg
E.Choking smell
F.Lamp black
J.contact process

SECTION B (70 Marks)
Answer all questions in this section

3. (a) Distinguish normal salts from acidic salts based on how they are formed.
(b) Give four uses of salts in daily life.
(c) Distinguish temporary hardness from permanent hardness of water.
(d) With the help of chemical equations, explain how you can remove each type of water hardness

4. (a) (i) Name any two (2) examples of industrial gaseous fuel.
(ii) List down four (4) environmental effects of using fire wood as source of energy.
(b) Give one reason for each of the following;
(i) Hydrogen gas is used in production of oxy-hydrogen flame.
(ii) Oxygen gas is collected over water during its preparation.
(iii) Water is referred as universal solvent.
(iv) Chlorine is added to water during urban water treatment.

5. (a)What do you understand by the following terms;
(i) Analyte
(ii) Stoichiometry
(b) An atom of element X having atomic number 11 combines with an atom of element Y having atomic number 9 to form a compound.

(C) Write the formula of the compound and state the type of bond formed in the compound.

(d) Give four properties of the compound formed in 5 (b)

6. (a) Briefly explain the three conditions for the chemical reaction to be at equilibrium-
(b In the industrial preparation of sulphur trioxide, equilibrium is established between sulphur dioxide and oxygen gas as follows:

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \quad \Delta H = -94.9 \text{ kJ/mol}$$

What will happen to the position of equilibrium if;
(i) Pressure of the system is increased

(ii) Temperature of the system is increased. (iii) Some amount of sulphur dioxide gas is removed.
(iv) Some amount of oxygen gas is added to the system.
7. (a) With the help of one example, give the meaning of the following;
(i) An electrolyte
(ii) A non- electrolyte
(b) The decomposition of an electrolyte by an electric current is called electrolysis. Name the ions that move to the;
(i) Negative electrode (cathode) during electrolysis
(ii) Positive electrode (anode) during electrolysis
(c) A solution of 1M copper (ii) chloride was electrolyzed using graphite electrodes. Write the reaction which took place at the;
(i) Anode
(ii) Cathode
8. (a) What do you understand by the following terms;
(i) Soil PH
(ii) Leaching
(ii.i) Soil fertility
(b) Explain any four (4) means in which a farmer can use so as to improve the fertility of the soil of his/her farm.
9. Assume you have visited a laboratory of Vikindu secondary school located in Pwani region and noted that in the school's laboratory the only materials available are;
Chemicals:
MANGANESE (IV) OXIDE
WATER AND POTASSIUM CHLORATE Apparatus:
RETORT STAND, DELIVERY TUBES, TEST TUBES, BEEHIVE STAND, BUNSEN BURNER, TROUGH AND RUBBER STOPPER.
(a) As a form four student, what can you prepare from the materials above?
(b) Draw a diagram to show how you can set your apparatus for that preparation
(c) Mention five (5) uses of the prepared item in (n) above.
10. (a) Write the structural formula of the following organic compounds;
(i) 2,4-dimethylhexane
(ii) 1,2-dichloropropane
(iii) Pent-2-ene
(b) Write any four isomers in pentene and for each isomer give its IUPAC name.

11. A piece of marble (calcium carbonate) was placed in a beaker containing an excess of dilute hydrochloric acid and standing on a direct reading balance. The mass of the marble and its contents was recorded in every 2 minutes.

Time minutes	2	4	6	8	10	12	14
Mass (g)	126.44	126.31	126.19	126.09	126.03	126.00	126.00

(a) Why there was a loss of mass?
(b) Write an equation for the reaction.
(c) State two (2) different ways in which the reaction could have been made more rapid.
(d) Why did the mass remain constant after 10 minutes?

12. (a) By using a balanced chemical equation, show how you can soften the following hard water,
(i) Permanent hard water by using washing soda.
(ii) Temporary hard water by boiling
(b) Write down any two (2) advantages of the following; (i) Hard water
(ii) Water treatment and purification
(iii) Change of state of matter

SECTION C (15 Marks)
Answer any one (I) question from this section.

13. (a) Iron and Aluminum both are found in similar oxides; Fe_20_3 and Al_20_3 but different methods for extraction are used. Explain why the methods for extraction are different.
(b) By using diagram describe the extraction of iron from the haematite ore by writing only the chemical equations for the reaction Involved in the extraction.
14. Explain three (3) causes of water pollution and suggest three (3) ways to prevent it

THE UNITED REUBLIC OF TANZANIA

SINGIDA REGIONAL MOCK EXAMINATION

CHEMISTRY 1 JULY ,2021

MARKING SCHEME

CHEMISTRY 1

FORM IV, 2021

I.	II.	III.	IV.	V.	VI.	VII.	VIII.	IX.	X.
B	A	C	C	C	B	A	E	C	D

2.

I.	II.	III.	IV.	V.
D	A	J	F	B

3. (a)Normal salt – Is a salt formed when all the replaced hydrogen ions of an acid are replaced by metal WHILE Acidic salt is a salt formed when only part of the replaced hydrogen ions per molecule of the acid is replaced.

(b) i/control soil PH

ii/used as an acid

iii/used as inorganic fertilizers

iv/used in preserving food

(c) Temporary hardness of water – is a property of water which can be softened by boiling. WHILE Permanent hardness of water – is a property of water which cannot softened by boiling.

(d)Temporary hard water can be removed by boiling which form carbonate.

Eg.Ca $(HCO_3)_2 \longrightarrow H_2O + CO_2 + CaCO_{3(s)}$.

Permanent hard water can be removed by adding sodium carbonate that form carbonate.

i.e $CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 + Na_2SO_4$

4.(a) i/Natural gas and coal gas.

ii/-Deforestation.

-Forest loss in the study areas.

-Soil erosion. (@ 0.5mark)

-Increase green house gases.

(b) i/Hydrogen can burn and gives high energy

ii/it is slightly soluble in water.

iii/It used to dissolve most various solute than any other solvent.

iv/to kill bacteria.

5.(a) i/Analyte – is a standardized solution or a solution added to the conical flask during Titration in which its concentration is not known. (1 mark)

ii/Stoichiometry – is a relationship between the relative quantities of substances taking part in chemical reaction and products formed. (1 mark)

(b) i/X+Y $\longrightarrow$ XY. (1mark)

ii/-soluble in water.

-Conduct electricity. (@ 0.5mark)

-Are usually crystalline

-Have high melting and boiling point.

6.(a) i/Must be reversible. (1mark)

ii/ Physical conditions must be constant

iii/Must be in a closed system

(b) i/Forward reaction will occur and hence more SO_3 will be formed.

ii/ Favours backward reaction and hence SO_2 and O_2 are more produced. (1mark)

iii/ Favours backward reaction to replace the removed SO_2

iv/ Favours forward reaction and hence more production of SO_3

7.(a) i/Any substance when in molten state can conduct electric current. Eg Nacl. (1.5mark)

ii/Any substance when in molten state cannot conduct electric current. Eg glucose. (1.5mark)

(b) i/Cation ii/Anion . (@0.5mark)

(c) i/$4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$ (2marks)

ii/$Cu^{2+} + 2e^- \longrightarrow Cu$ (2marks)

8.(a) i/Measure of acidity or alkalinity of the soil.

ii/Process in which nutrients are carried to lower levels far from plant roots. (@1mark)

iii/Is the ability of soil to sustain plant growth by providing essential plant nutrients.

(b) i/Crop rotation.

- Growing various types of crops on the same field.

ii/Contour farming (@1mark)

-Cultivation across the slope.

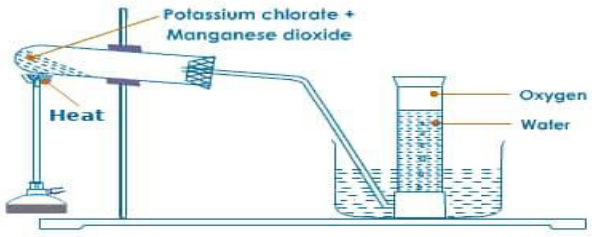
iii/Using manure

iv/Controlled grazing.

-Keeping the right number of animal on a piece of land prevents overgrazing.

9.(a)Oxygen gas. (1mark)

(b)



(3MARKS)

(c) i/Used for living organisms.

ii/Used in hospital. (@ 1mark)

iii/Used in manufacturing of chemical.

iv/Used in welding.

10.(a) i/ CH_3- CH- CH_2- CH- CH_2-CH_3

CH_3 CH_3

ii/CH_3- CH- CH_2CL (@ 1mark)

Cl

iii/CH_3-CH= CH- CH_2- CH_3

(b) i/CH_3- CH- CH = CH_2.

Ch_3 3- methyl but – 1- ene

ii/$CH_3CH_2C=CH_2$

2 – methyl but – 1 – ene (@2marks)

iii/$CH_3C=CHCH_3$

2 – methyl but -2-ene

iv/$CH_3CH_2CH=CHCH_3$

Pent – 2- ene.

11.(a)Calcium carbonate decrease by reaction with Hydrochloric acid. (1mark)

(b)$CaCO_3$ + HCl $\longrightarrow$ $CaCl_2 + H_2O + CO_2$. (2marks)

(c) i/Reduce surface area of calcium carbonate.

Ii/Increase concentration of HCL acid. (@ 1 mark)

(d) Reaction is completed. (1mark)

12.(a) i/$CaSO_{4(aq)} + Na_2CO_{3(aq)}$ $\longrightarrow$ $CaCO_{3(s)} + Na_2SO_{4(aq)}$

ii/$Ca(HCO_3)_{2(aq)}$ $\longrightarrow$ $CaCO_{3(s)} + CO_{2(g)} + H_2O_{(l)}$ (@ 2marks)

(b) i/- It has minerals that promote teeth and bones.

- It promotes animal shells. (@ 1mark)

ii/- It kills bacteria.

-It ensures clean and safe water.

iii/- Water cycle.

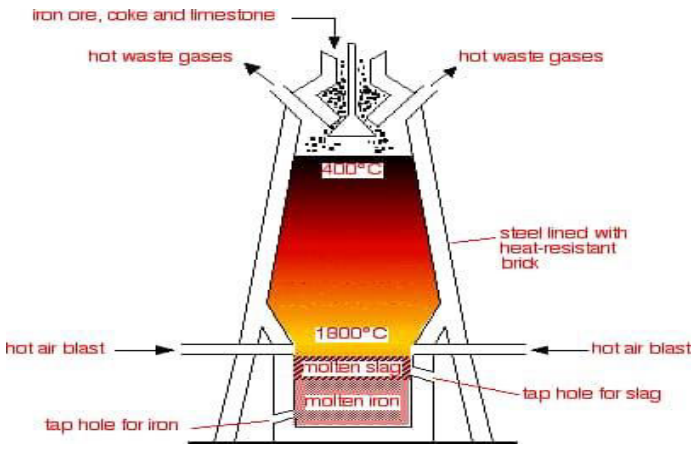
-It helps in refrigeration.

SECTION: C (15marks)

13.(a)They have different reactivity. (01 mark)

(b)Iron can be extracted from hematite over by putting/adding the over with limestone and coke into blast furnace. (01 mark)

(Drawn)



(3 marks)

-Chemical equations involved:

I/coke (c) burns in oxygen.

i.e $C_{(2)} + O_2 \longrightarrow CO_{2(G)}$

ii/$CO_{2(g)}$ formed react with more coke.

$CO_{2(g)} + C_{(s)} \longrightarrow 2CO_{(g)}$

Iii/carbon monoxide formed reduces iron III oxide to spongy iron.

i.e $Fe_2O_3 + 3CO_{(g)} \longrightarrow 2Fe_{(l)} + 3CO_{2(g)}$

(iv) Limestone decomposes at 600 C to quicky lime (cao)

i.e $CaCO_{3(s)} \xrightarrow{600^{\circ} C} CaO + CO_2$

(v)Cao react with earth impurities in the ove (siO_2) to form slag

i.e $Cao_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3(l)}$ (02Marks @)

14.Introduction (03marks)

Causes of water pollution;

i/ Pesticides

-It drops onto the soil and carried into local streams and rivers by surface run off.

Ii/Industries process

Iii/Fertilizers

Iv/Mining

V/Oil spills. (02marks@)

.Ways to prevent water pollution

I/reducing and pesticide pollution.

Ii/Treating sewage and industrial waste

Iii/Reducing mercury emissions.

Iv/Cleaning up existing and abandoned mines.

V/Reducing pollution from oil spills.

Vi/Controlling coastal development. (02marks@)